

## ***Guardians of the Ancient Treasure Hunt***

## ***Immersive Technical Manual***

### ***For the Guardians of the Ancient Treasure Hunt***



#### **Prologue – The Chrono-Shatter**

Long ago, the **Temporal Stabilization Key (TSK)** was shattered, its fragments sealed within eight chambers representing eight technological eras. You, the **Temple Masters**, are the guardians of these chambers. Explorers (**Chrono-Agents**) will travel through 8 stations. Each station is a self-contained embedded system puzzle combining **sensors, actuators, and logic**—and only when solved correctly will agents receive the relic (clue) needed for the next chamber.



#### ***Now Let's go to the Chamber Descriptions***

Note: In all station, there must be a way to reset the system and returns it to the initial state/positions, etc.

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### Station 1: The Sun Dial Lock - The Ancient Temple

#### **Story Overview:**

The Agents enter an ancient, silent chamber. A heavy stone dial on a pedestal must be physically aligned to a peak light intensity, while an artifact must be placed at a precise distance to satisfy the "Silent Echo."

#### **Required Clue:**

The statement *"Open sesame"*

#### **Subsystems:**

1. **Enabling Mechanism (Sequential/AI):** Microphone listens for the target keyword ("Open sesame") to activate the remaining subsystems.
2. **Light Alignment Chamber (Parallel):** Monitors light intensity via LDR.
3. **Acoustic Chamber (Parallel):** Ultrasonic sensor measures the distance to a placed artifact.
4. **Relay Lock Mechanism (Sequential):** Waits for the two parallel alignment flags to grant the reward.

| Feature         | Details                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------|
| <b>Inputs</b>   | <b>LDR (Analog)</b> , Ultrasonic Sensor, Microphone                                                   |
| <b>Outputs</b>  | <b>DC Motor</b> , LCD, Piezo Buzzer.                                                                  |
| <b>AI Twist</b> | <b>Voice Recognition (TinyML)</b> to recognize the opening statement and enable all other subsystems. |

#### **Overall Logic:**

- The room is initially dormant. The player must speak the opening statement: **"Open sesame."**
- **If Incorrect:** The **TinyML model** fails to recognize the phrase. The **Buzzer** emits three short error beeps.
- **If Correct:** The **TinyML model** recognizes the phrase. The **Buzzer** emits a long success tone. The **DC Motor** rotates 90° to open a small panel, revealing the **LCD Screen**. The LCD writes: *"The room has been activated successfully, start playing to get your next clue."*
- The player must then manipulate the environment, holding a flashlight towards the **LDR (Analog)** and adjusting the distance of an object from the **Ultrasonic Sensor**.
- The system waits for two parallel flags:
  - **Light\_Aligned:** LDR reading is in target window (e.g., 100 LUX  $\pm$  5).
  - **Acoustic\_Aligned:** Ultrasonic reading is in target window (e.g., 70 cm  $\pm$  2 cm).
- **When both are TRUE:** The **LCD** immediately writes: *"Congratulations! You solved the room and your next clue is 7536."*

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**Gained Reward:**

A new **4-Digit Code** (e.g., 7536).

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### Station 2: The Neural Calibrator - The Chamber of the Clay Golem

#### **Story Overview:**

Agents must interface with a networked mechanical clay hand (**The Golem's Gripper**) to retrieve a delicate **Chrono-Gem (the jewel)** without shattering it. The Golem's hand holds a container that must be squeezed with precise force. Once the force is correct, a smaller **QR-encoded Data Chip (the chip)** is released. This requires controlling the **Servo Motor** (which drives the gripper) precisely via an **Engraved Control Tablet** (web interface).

#### **Required Clue:**

**4-Digit Code** (from Station 1).

#### **Subsystems:**

1. **Network Activation (Sequential):** Keypad validates the 4-Digit Code and initializes the Pico W's web server.
2. **Remote Grip Control (Parallel):** The web server UI allows the agent to control the Servo Motor (Golem's hand) angle using directional inputs.
3. **Force Feedback Monitor (Parallel/AI):** Measures grip strength via FSR and streams the live data to the web interface.
4. **Reward Release Mechanism (Sequential):** DC Motor executes the final chip release command upon success.

| Feature                       | Details                                                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (3+; 1 Analog):</b> | 4x4 Keypad, <b>Force Sensing Resistor (FSR) (Analog)</b> (Pressure Sensitive Gem Mount) inside gripper, Web Browser Screen ( <b>Directional Control Arrows</b> via Pico W Wi-Fi, Start Button).        |
| <b>Outputs (3+):</b>          | <b>Servo Motor</b> (Golem's Clay Hand <b>Gripper Actuator</b> ), <b>DC Motor</b> (Chip Release Mechanism), LCD, RGB LED (Status).                                                                      |
| <b>AI Twist:</b>              | <b>Web Server Interface with Data Visualization:</b> The FSR's analog reading is streamed live via WebSockets and displays as a real-time " <b>Grip Integrity Meter</b> " on the Agent's phone screen. |

#### **Overall Logic:**

- The player enters the **4-Digit Code** on the Keypad.
- **If Correct:** The Pico W activates its Wi-Fi module, the **RGB LED turns Yellow**, and the LCD displays the web server IP address (for the Control Tablet).
- The Agent connects to the Wi-Fi and accesses the control page, seeing the live **Grip Integrity Meter**.
- The Agent presses the **Start Button** on the web interface. The **RGB LED turns Red** (Adjustment Mode).
- The Agent uses the web interface's **Directional Arrows (Close/Open)** to command the **Servo Motor** to slowly close the Golem's hand onto the Jewel/Gem.

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- The Agent must stop pressing the controls when the meter enters the target zone (e.g.,  $180 \pm 5$  units).
- **If Target Reached:** The **RGB LED turns Blue** (Hold Mode). A 5-second success timer begins.
- **If SUCCESS (Blue held for 5 seconds):** The **RGB LED turns Green**. The dedicated **DC Motor rotates CW** (Clockwise) to actuate the reward chute, releasing the **Chip with ArUco Marker printed on it**.
- **If FAILURE (Pressure drops/exceeds tolerance during Hold Mode):** The **RGB LED flashes Red/Yellow** (Error Cue). The dedicated **DC Motor rotates CCW for 5 seconds** (a failure sound/reset cue). The system resets, and the Servo opens fully to release the Jewel and “*TRY Again*” will appear on the LCD.



### **Gained Reward:**

An **ArUco Marker** (printed on the released chip).

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### Station 3: The Verification Echo - The Crypt of the Ancestral Voice

#### Story Overview:

A sarcophagus housing the next clue is sealed by a spectral ward. The Agents must present the **ArUco Marker** obtained from Station II to an **Ancient Ocular Stone (Camera)** for precise, real-time visual verification. The marker must be presented at a specific distance and orientation to unlock the sarcophagus.

#### Required Clue:

The **ArUco Marker** (from Station 2).

#### Subsystems:

1. **Presence Detection (Parallel):** PIR detects agent presence to set the initial ready state.
2. **Scanning Activation (Sequential):** Tactile Sensor/Pad (The Ancient Orb) must be continuously pressed to enable the camera/AI subsystem.
3. **Visual Verification (Parallel/AI):** Camera Module recognizes the ArUco marker ID and checks its orientation and distance/size in the frame.
4. **Motorized Access Mechanism (Sequential):** DC Motor controls the physical lock securing the relic by **lifting a heavy stone block**.

| Feature                          | Details                                                                                                                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (3+; 1 Analog):</b>    | <b>Small Camera Module</b> (The Ancient Ocular Stone), PIR Motion Sensor, <b>Tactile Sensor/Pad</b> (The Initiation Orb).                                                                                                      |
| <b>Outputs (4+; 1 DC Motor):</b> | <b>DC Motor</b> (Sarcophagus Lid Actuator/Stone Block Lift), 16x2 LCD Screen, <b>RGB Status LED (Integrated into the Orb)</b> , <b>Piezo Buzzer</b> .                                                                          |
| <b>AI Twist:</b>                 | <b>Computer Vision (ArUco Marker Detection):</b> The Camera Module detects a specific <b>ArUco Marker ID</b> and measures its real-time spatial properties (orientation, distance, scale) to ensure it is presented correctly. |

#### Overall Logic:

- **PIR** detects the Agent; the system awakens. The **LCD** prompts: *"Place Hand on the Orb and Present the Seal."*
- **Hand on Orb Check (Continuous):** The system continuously monitors the **Tactile Sensor/Pad**.
  - **If Hand Detected:** The **RGB LED (Orb)** glows **Blue**. The **Camera Module** begins searching for the marker.
  - **If Hand Removed:** The **RGB LED (Orb)** flashes **Red** and the **Piezo Buzzer** sounds an error alarm. The visual verification halts until the hand is returned.
- **While Hand is on the Orb (Blue Status):**
  - The Agent must hold the **ArUco Marker** in front of the **Camera Module** and rotate it until the required position is detected.

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- **If Visual Check SUCCESS:** If the **Computer Vision Model** detects the correct **ArUco Marker ID** and confirms its orientation (e.g.,  $0^\circ \pm 5^\circ$  tilt) and distance (e.g.,  $15 \text{ cm} \pm 2 \text{ cm}$ ) are correct for 5 seconds, the **RGB LED (Orb)** turns solid **Green**. The **DC Motor** rotates  $90^\circ$  to unlock and **lift the stone block** smoothly, revealing the reward.
- **If Visual Check FAILURE (Incorrect Position/Marker):** The **Piezo Buzzer** emits a harsh denial tone while the Agent tries to adjust the marker. The **LCD** displays: *"Verification Failed. Adjust Seal and Realign."*



### **Gained Reward:**

An **RFID Tag** (embedded in a lab access card).

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### Station 4: The Abyssal Balance - The Flooded Treasury of the River God

#### **Story Overview:**

The Agents must stabilize a miniature ballast tank (representing a water-filled treasury chamber) to match the precise pressure setting encoded on the RFID tag. Manual adjustment of a brass valve controls the water level in the tank.

#### **Required Clue:**

The **RFID Tag** (from Station 3). The RFID card contains a target hydrostatic pressure value (e.g., 750).

#### **Subsystems:**

1. **RFID Read & Target Setting (Sequential):** RFID reader verifies the tag and loads the target level into the system memory.
2. **Level Control Actuation (Parallel):** Potentiometer input drives the pump speed and direction.
3. **Hydrostatic Monitoring (Parallel):** Analog Water Level Sensor measures the fluid height in real time.
4. **Reward Display & OCR (Sequential/AI):** Activates the UV LED and uses the Camera to read the revealed text.

| Feature                       | Details                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (4+; 1 Analog):</b> | RFID Reader, <b>Analog Water Level Sensor (Analog)</b> (Pressure Stone Gauge), Potentiometer (Brass Valve Handle), Push Button (Submit/Confirm), <b>Small Camera Module</b> (OCR Scanner).                                                                                                                                      |
| <b>Outputs (2+; 1 Pump):</b>  | <b>Electrical Pump</b> (Controls water flow <b>In and Out</b> ), Small LCD Display (shows current level), UV LED (Reward Activation).                                                                                                                                                                                           |
| <b>AI Twist:</b>              | <b>Computer Vision (Optical Character Recognition - OCR):</b> Once the correct water level is achieved, the UV LED lights up a hidden text area. The <b>Camera Module</b> captures this UV-revealed text and uses <b>OCR (Optical Character Recognition)</b> to read the clue, displaying the final sequence on the LCD screen. |

#### **Overall Logic:**

- Agent scans the **RFID Tag**. **SUCCESS:** LCD displays "*Target Level: 750 units.*" and the **Small LCD** constantly updates the current reading from the **Analog Water Level Sensor**.
- Agent turns the **Potentiometer (Analog)** control dial. This input is directly mapped to the **Electrical Pump**'s speed and direction: turning clockwise pumps water **IN** (level rises), and turning counter-clockwise pumps water **OUT** (level drops). This allows the Agent precise, bidirectional control.
- The Agent tries to manually stabilize the level near 750.

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- Agent presses the **Submit** button to check the state.
- The **Water Level Sensor** reading is displayed on the LCD once the Submit button is pressed.
- **If SUCCESS:** If the **Analog Water Level Sensor** reading is within the target tolerance (e.g.,  $750 \pm 3$  units):
  - A hidden **UV LED** lights up, revealing text (e.g., "MIX 9:5:2").
  - The **Camera Module (OCR Scanner)** automatically reads this revealed text.
  - The **Small LCD** displays the captured text: "*Clue Captured: MIX 9:5:2*".
- **If FAILURE:** The **Electrical Pump** quickly cycles water out, forcing the agent to restart the fine-tuning process.

### **Gained Reward:**

A **UV Light Text** (revealed by the UV LED on the station).

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### Station 5: The Primordial Mix - The Chamber of Celestial Pigments

#### **Story Overview:**

The Agents must interface with a **Spectral Calibration Interface (Web UI)** to digitally tune the intensity of the three primary light vessels (R, G, B LEDs). The system uses a Wi-Fi connection to stream **real-time visual feedback** on the mixed color and target color to the Agent's screen, validating the mix using the target spectral ratio.

#### **Required Clue:**

The **UV Light Text** (from Station 4). This text provides the target spectral ratio (e.g., "Mix R:9, G:5, B:2").

#### **Subsystems:**

1. **Network Activation (Sequential):** Pico W initializes the web server and provides the IP for the interface.
2. **Spectral Flow Interface (Parallel):** Three Potentiometers measure the agent's desired R, G, B input and stream this data to the web interface.
3. **Chromatic Validation System (Parallel/AI):** The single **TCS3200 Color Sensor** reads the combined light output and feeds the data to the clustering algorithm.
4. **Proportional Reward Actuator (Sequential):** DC Motor rotates the ceremonial cauldron cover based on the calculated correctness percentage.

| Feature                          | Details                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (4+; 1 Analog):</b>    | <b>TCS3200 Color Sensor (Analog)</b> (Sky-Stone Spectral Reader), 3x Potentiometers (Star-Vessel Flow Regulators), Push Button (Scan), <b>Web Browser Input</b> (Calibration Interface via Pico W Wi-Fi).                                                                                                                                          |
| <b>Outputs (4+; 1 DC Motor):</b> | <b>DC Motor</b> (Ceremonial Cauldron Cover), 3x High-Intensity LEDs (R, G, B), <b>RP2040 Wi-Fi Module</b> (Web Server Host), 16x2 LCD Display (for IP address).                                                                                                                                                                                    |
| <b>AI Twist:</b>                 | <b>Web Server Interface with Data Visualization (Real-time Color Feedback):</b> The Pico W hosts a web server that streams the current color measured by the TCS3200 (Current Mix) via WebSockets. The interface displays a visual swatch of the <b>Current Mix</b> alongside the <b>Target Color</b> and the live <b>Correctness Percentage</b> . |

#### **Overall Logic:**

- The **RP2040 Wi-Fi Module** initializes the network and displays the web server **IP address** on the **16x2 LCD Display** upon pressing the **start button**.
- Agents connect to the Wi-Fi and open the **Spectral Calibration Interface** in their browser.
- Agents adjust the three **Potentiometers (Analog inputs)**. The Pico W reads these values and sends the corresponding PWM signals to change the brightness of the R, G, and B LEDs below the sensor.

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- The **LCD** continuously cycles to display the **Potentiometer raw input values** (0-1023) for R, G, and B, acting as a *local, quantitative guide* for the agents to tune the dials.
- The **Potentiometers readings** are mapped to the colors percentages which are sent to the web server that calculates **Current Mixed Color** and display it on the **Web Interface**.
- The **Web Interface** updates in real-time, showing: a swatch of the **Target Color**, a swatch of the **Current Mixed Color**, and the live **Correctness Percentage** (0-100%)
- The **TCS3200 Color Sensor** continuously measures the **Current Mixed Color** shown on the web interface and update the live **Correctness Percentage** (0-100%).
- Meanwhile, the **DC Motor** rotates the **Ceremonial Cauldron Cover** to an angle **proportional** to the live **Correctness Percentage**, providing immediate, tangible physical feedback.
- **Validation and Reward:**
  - **If SUCCESS:** If the calculated **Correctness Percentage** is within the acceptance correctness range  $\geq 97\%$ , the motor completes the full rotation, fully revealing the reward. The web interface displays: *"Composition Validated."*
  - **If FAILURE ( $< 97\%$ ):** The motor holds its current position.

### **Gained Reward:**

A **Geometric Sequence** (e.g., 2, 6, 18, 54, X).

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### Station 6: Mimic's Reflection - The Smart Mirror of the Chronos

#### Story Overview:

Agents must translate the final number of the geometric sequence into a specific, sustained sequence of facial gestures. They enter the calculated number on a keypad. A **Smart Mirror** then displays the 3 required gestures sequentially. The Agents must perform each gesture, which is verified in real-time by a **Camera/AI model** before the next gesture prompt appears. Success rotates the mirror to reveal the next clue.

#### Required Clue:

A **Geometric Sequence** (from Station 5). The numerical value of the next term in the sequence (e.g., 2, 6, 18, 54, **162**) is the key.

#### Subsystems:

1. **Numeric Input & Gesture Mapping (Sequential):** Keypad/Touchscreen validates the calculated number and maps the 3 digits to 3 target gestures.
2. **Real-time Facial Recognition (Parallel/AI):** Camera Module uses Computer Vision to verify the user's expression against the target.
3. **Smart Mirror Feedback (Parallel):** Integrated Display provides instruction, countdown, and confirmation messages.
4. **Reward Release (Sequential):** Stepper Motor rotates the mirror panel to display the hidden clue.

| Feature                         | Details                                                                                                                                                                                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (3+; 1 Camera):</b>   | 4x4 Keypad/Touchscreen (Number Input), <b>Small Camera Module</b> (Facial Recognition/Smart Mirror), Push Button (Start Sequence).                                                                                                                             |
| <b>Outputs (3+; 1 Stepper):</b> | <b>DC Motor</b> (Mirror Rotation Actuator), <b>LCD/Integrated Display</b> (Smart Mirror Text Feedback), RGB Status LED.                                                                                                                                        |
| <b>AI Twist:</b>                | <b>Computer Vision (Facial Gesture Recognition):</b> A model trained on facial landmarks recognizes specific, discrete expressions (mapped 0-9) in real-time. It must classify the user's face as the target gesture (e.g., 'Smiling') for a sustained period. |

#### Gestures Mapping (0-9):

| Digit | Gesture              | LCD Prompt Example                                  |
|-------|----------------------|-----------------------------------------------------|
| 0     | Neutral/Relaxed Face | <i>Hold a calm, centered expression.</i>            |
| 1     | Smile/Laughing       | <i>Show the Chronos your joy! Keep smiling!</i>     |
| 2     | Right Eye Blink/Wink | <i>A secret signal. Blink your right eye once.</i>  |
| 3     | Sad/Frown            | <i>Express the weight of time.</i>                  |
| 4     | Raise Right Eyebrow  | <i>Question the timeline. Lift your right brow.</i> |
| 5     | Open Mouth (O-Shape) | <i>The shape of the original void.</i>              |
| 6     | Surprise (Wide Eyes) | <i>Show your astonishment! Widen your eyes.</i>     |
| 7     | Pout/Duck Face       | <i>Show your displeasure with the lock.</i>         |
| 8     | Left Eye Blink/Wink  | <i>A subtle clue. Blink your left eye once.</i>     |

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|   |                  |                                      |
|---|------------------|--------------------------------------|
| 9 | Right Cheek Puff | Inflate the right cheek for balance. |
|---|------------------|--------------------------------------|

### **Overall Logic:**

- Agent calculates and enters the 3-digit number (e.g., **162**) on the **Keypad/Touchscreen**.
- **If Correct Code (162):** The **RGB LED turns Blue**. The **Integrated Display (Mirror)** prompts: *"Sequence Unlocked. Stand before the Mirror and press START."*
- Agent presses the **Start Button**. The **RGB LED turns Red** (Active Scan).
- **Gesture Sequence (3 Stages):** The system cycles through the three digits one by one.
  1. **Stage 1 (Digit 1):** The mirror displays the first target gesture (e.g., Digit 1 → Smile) and starts a 10-second countdown. The **Camera Module** continuously monitors the face.
    - **Real-time Feedback:** If the **AI model** recognizes the correct gesture, the **Integrated Display** shows: *"Gesture Recognized. Hold it!"* If incorrect, it shows coaching text from the table above (e.g., *"Keep smiling!"*).
    - **If SUCCESS:** Correct gesture held for 10 consecutive seconds. Moves to Stage 2.
    - **If FAILURE:** If time runs out, the **RGB LED flashes Red**, and the sequence resets to Stage 1, displaying an error message.
  2. **Stage 2 (Digit 2):** Same logic applies for the second digit (e.g., Digit 6 → Surprise).
  3. **Stage 3 (Digit 3):** Same logic applies for the third digit (e.g., Digit 2 → Right Eye Blink).
- **Final Reward Actuation:**
  - **If all 3 Stages are SUCCESSFUL:** The **RGB LED turns Green**. The **Stepper Motor** rotates the mirror 180° displaying the physical clue. The **Integrated Display** shows: *"Reflection Verified. Chronos Opens."*

### **Gained Reward:**

A **Melody Pattern** (a sequence of notes: C4, G4, A4, F4...).

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### Station 7: The Astrolabe of the Heavens - The Celestial Lock

#### **Story Overview:**

The ultimate test of fine physical control and spatial reasoning. Seekers must first input a specific sonic frequency pattern—**The Call of the Oracle**—by striking the **Inscribed Tone Panel (Capacitive Touch Keys)**. Once activated, they use a **Wearable Sensor Gauntlet (The Inertial Controller)** to physically align a **3-Axis Gimbal-mounted Satellite Cube** (marked with an Aruco target) into a specific 3D orientation. The system continuously displays the required angle on the **Local Guide (LCD)**, while a **High-Resolution Scanner (Camera)** silently monitors the cube. Perfect alignment, verified by the computer vision system, triggers a complex, multi-step **DC Motor Safe Key Sequence** and releases the final scroll via a magnetic lock.

#### **Required Clue:**

The **Melody Pattern** (from Chamber VI). Tapping this precise pattern of notes/frequencies is the sonic key that activates the control screen.

#### **Subsystems:**

1. **Melody Activation (Sequential):** A set of **Capacitive Touch Keys** are arranged on a stone panel. The system listens for the correct sequence of inputs, validating both the correct key (pitch) and the precise temporal spacing between key strikes.
2. **Inertial Controller (Parallel):** The **MPU-6050 Accelerometer/Gyroscope (Analog)** is housed in the Gauntlet. It translates the physical 3D tilt/rotation (Yaw, Pitch, and Roll) of the seeker's hand into real-time angular commands.
3. **Physical Alignment Actuator (Parallel): Three Servo Motors (3-Axis Gimbal)** receive the MPU-6050 data and directly mirror the Seeker's hand motion, positioning the **Aruco-marked Satellite Cube** in three dimensions.
4. **Visual Verification (Parallel/AI):** The **Small Camera Module** continuously scans the Aruco marker on the Satellite Cube, determining its real-time 3D pose (position and orientation) relative to the target solution zone.

**Clue Reveal (Sequential): A powerful DC Motor executes a predefined, timed, and varied sequence of speeds and directions to act as a Safe Key, which, upon completion, disengages the Magnetic Lock.**

| Feature                        | Details                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs (3+; 2 Analog):</b>  | <b>MPU-6050 Accelerometer/Gyroscope (Analog)</b> (Gauntlet of the Navigator), <b>4x Capacitive Touch Keys</b> (The Inscribed Tone Panel), <b>Small Camera Module</b> (High-Resolution Scanner for Aruco Pose).                                                                                           |
| <b>Outputs (6+; 4 Motors):</b> | <b>3x Servo Motors</b> (Yaw, Pitch, Roll Actuators for the 3D Gimbal), <b>DC Motor (Safe Key Actuator)</b> , <b>Electromagnet</b> (Magnetic Door Lock Release), <b>16x2 LCD Display (Local Guide)</b> , continuously showing Target Angles), Status LEDs (Ready/Solved State), Piezo Buzzer (Error Cue). |
| <b>AI Twist:</b>               | <b>Computer Vision (Aruco Pose Verification):</b> The model detects the                                                                                                                                                                                                                                  |

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|  |                                                                                                                                                                                                       |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | specific ArUco marker ID and must verify its exact 3D pose (position, rotation, and scale) within tight, multi-axis tolerances for <b>2 consecutive seconds</b> before signaling the unlock sequence. |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### **Overall Logic:**

- **Initial State:** The system is dormant. Status LEDs are Red. The **16x2 LCD Display** is blank. The **Electromagnet** is **ON**, locking the reward door.
- **Melody Activation (Sonic Key Check):** The player must physically tap the **Inscribed Tone Panel (Capacitive Touch Keys)** in the precise sequence and timing (C4, G4, A4, F4...).
  - **If Correct Sequence Detected:** The system initializes. The **Status LEDs** turn solid Green. The **16x2 LCD Display** activates, displaying: "*Celestial Lock Active. Align Cube. Yaw: 90 Pitch: 90 Roll: 90.*" (The target angles).
  - **If Incorrect:** The **Piezo Buzzer** emits a distorted, grating error tone, and the system resets the input attempt timer.
- **Control Phase:** The Seeker uses the **MPU-6050 Gauntlet** to physically tilt and rotate their hand sequentially (yaw then pitch then roll).
  - **Physical Mirroring:** The MPU-6050's stabilized 3D orientation (Yaw, Pitch, Roll) is used to drive the **Three Servo Motors**, which move the **Satellite Cube's** gimbal.
  - **Local Feedback:** The **LCD** continuously displays the **real-time MPU-6050 orientation data** (e.g., "Yaw: 95 Pitch: 88 Roll: 91").
- **Vision Check Loop:** The **Camera Module** constantly checks the Aruco marker on the cube against the predefined target position and orientation (e.g., Yaw 90°, Pitch 90°, Roll 90°).
- **If Vision Check SUCCESS (Target Pose Verified):** If the **Computer Vision Model** confirms the precise 3D pose for **2 consecutive seconds**, the system locks the successful state. The **Status LEDs** flash Blue rapidly. The **LCD** displays: "*Target Locked: Initiating Key Sequence.*"
- **Reward Actuation (Safe Key Sequence & Magnetic Release):** The DC Motor immediately executes the predefined sequence:
  1. **CW Rotation, Fast (PWM 255)** for 1.5 seconds (High-speed whirl).
  2. **Pause** for 0.5 seconds.
  3. **CCW Rotation, Half Speed (PWM 128)** for 2.5 seconds (Slow, grinding unlock).
  4. **Stop.**

*Upon successful completion of the DC Motor sequence:* The system immediately de-energizes the **Electromagnet** (Magnetic Lock → OFF). The magnetic seal is broken, and the final scroll is released (e.g., spring-loaded door opens).
- **If Vision Check FAILURE (Pose Lost):** The **LCD** displays "Align Cube: Error," and the Seeker must re-acquire the target pose.

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### **Gained Reward:**

**A Set of Four Celestial Coordinates (e.g.,  $A = 7, B = 5, C = 3, D = 6$ ) - The Celestial Alignment Key.**

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### Station 8: The Chronos Synthesis Vault - The Heart of the Forgotten God

#### **Story Overview:**

You stand before the final lock, the **Chronos Synthesis Vault**. It requires two distinct stages of verification to secure the treasure: **Digital Synchronization** and **Pattern Synthesis**. First, Agents must use the collected numerical fragments to unlock a **Quantum Matrix Interface** on the web, where they must precisely control their physical distance from a **Proximity Sensor** to stabilize a digital maze. Completing this digital task reveals the final Master Code digit. Finally, they must synthesize the complete nine-digit sequence into a physical pattern—the **Chronos Knot**—and present it to the **Camera Module** for biometric pattern verification, triggering the vault's opening.

#### **Required Clue:**

The Celestial Alignment Key (from Chamber VII):  $A = 7, B = 5, C = 3, D = 6$ .

#### **Subsystems:**

1. **Code Activation (Sequential):** The web interface's virtual keypad accepts the four-digit key ( $A, B, C, D$ ). It successfully validates the code and initiates the digital maze challenge.
2. **Web Matrix Interface (Parallel):** The Pico W-hosted Wi-Fi server. It runs the continuous maze game and displays the final digit **4** upon completion of the digital synchronization phase.
3. **Signal Stabilization (Parallel):** The Ultrasonic/Proximity Sensor array. It continuously reads the player's distance, calculating and modulating the 'Jitter' (difficulty) of the Web Matrix Interface in real-time.
4. **Pattern Scanner (Sequential/AI):** The Camera Module and Computer Vision Model. It is triggered after the full code assembly and performs a critical one-time analysis of the drawn Chronos Knot pattern to verify the final solution.
5. **Vault Actuator (Sequential):** The DC Motor subsystem. It runs the linear stage mechanism to physically present the final treasure, marking successful mission completion.

| Feature  | Details                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs   | <b>Ultrasonic/Proximity Sensor</b> (The Jitter Modifier), <b>Camera Module</b> (Pattern Recognition).                                                                                                                                                                       |
| Outputs  | <b>Large LCD/Touchscreen</b> (Feedback), <b>Pico W Wi-Fi</b> (Hosts Web Maze Game), <b>DC Motor</b> (Linear Stage Actuator), <b>RGB Status LED</b> , <b>Piezo Buzzer</b> .                                                                                                  |
| AI Twist | <b>Computer Vision (Pattern Recognition):</b> The Camera Module analyzes the image for the precise 'Chronos Knot' pattern drawn on the number grid ( $7 \rightarrow 5 \rightarrow 3 \rightarrow 6 \rightarrow 9 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 7$ ). |

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### **Overall Logic:**

The challenge is executed in two distinct phases:

#### 1. Phase 1: Digital Synchronization

- **Access:** Agent accesses the web interface. The web page displays a **4x4 virtual keypad** and a "START" button. Agent enters the 4-digit sequence 7536 on the virtual keypad.
- **Web Game Launch:** On correct code entry, the local **RGB Status LED** turns **Blue**, and the web page transitions to the grid maze interface with the **on-screen joystick controls** (Up, Down, Left, Right).
- **Jitter Mechanic:** Agent connects to the web game and begins the maze. The **Proximity Sensor** continuously reads distance, which applies 'Jitter' to the game cursor if the player moves outside the optimal range ( $20\text{ cm} \pm 5\text{ cm}$ ).
- **Clue Reveal:** Successfully navigating the maze clears the Jitter and the web interface reveals the 8th Master Code Digit: **4**.

#### 2. Phase 2: Pattern Synthesis

- **Deduction:** Agents assemble the full nine-digit code: **7 5 3 6 9 5 1 4 7**. Note that the last 9th digit is simply the same as the first digit to close the shape.
- **Drawing & Scan:** The Agent draws the **Chronos Knot** (the infinity-like shape formed by the sequence on a number grid) and places it under the **Camera Module**.
- **Final Verification:** The **Computer Vision Model** analyzes the drawing.
- **If Pattern SUCCESS:** The **RGB Status LED** turns **Solid Green**. The **DC Motor** activates the linear stage, moving the cart with the final treasure box.
- **If Pattern FAILURE:** The Piezo Buzzer emits an error cue, and the system prompts a retry.

### **Gained Reward:**

The activation of the **DC Motor** and linear stage, delivering the final Treasure Chest and signaling Mission Success.